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# Solar PV proposal for Sample client (landed home)

Taman Example, Kuala Lumpur

A **3.51 kWp** rooftop solar system is expected to produce about **4,845 kWh** in its first year, covering **33%** of your electricity use. Under TNB domestic tariff (from 1 July 2025) with Solar ATAP export credit it should cut your bills by about **RM 2,382** in year one and pay for itself in about **5.2 years**.

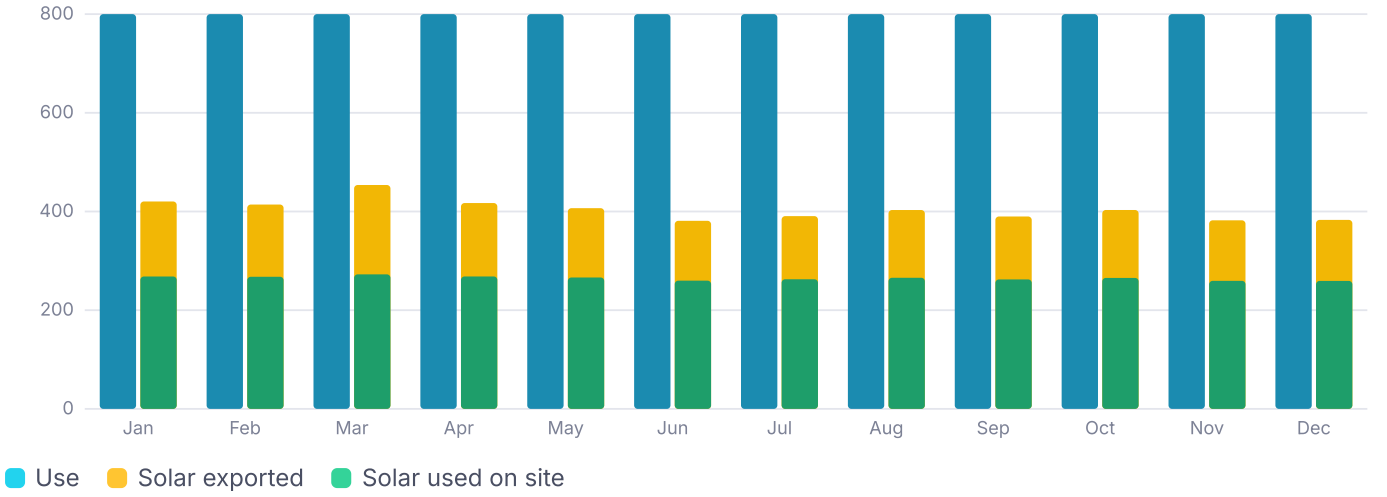
| | |
|---|---|
| **3.51 kWp**<br>System size | **RM 12,285**<br>Investment |
| **RM 2,382**<br>Savings, year 1 | **5.2 yrs**<br>Payback |
| **4,845**<br>kWh a year | **18.7%**<br>Return (IRR) |
| **RM 47,292**<br>Net gain, 25 years | **3.7 t**<br>$CO_2$ avoided a year |

## 1. Proposed system

| | |
|---|---|
| Modules | 6 × Generic 585 W n-type TOPCon (144 half-cell, 182 mm) |
| Inverter | 1 × Generic 3 kW single-phase (1 MPPT) |
| DC / AC | 3.51 kWp / 3.0 kWac |
| Strings | 1 × 6 modules |
| Orientation | 10° tilt, facing south (180°) |
| Roof area | ≈ 18 m² |

| | |
|---|---|
| Location | 3.139, 101.687 |
| Sunshine | 1,751 kWh/m² a year (4.80 peak sun hours a day) |
| Specific yield | 1,380 kWh/kWp |
| Performance ratio | 79.4 % |
| Your use | 800 kWh a month |
| Solar used on site | 66% |

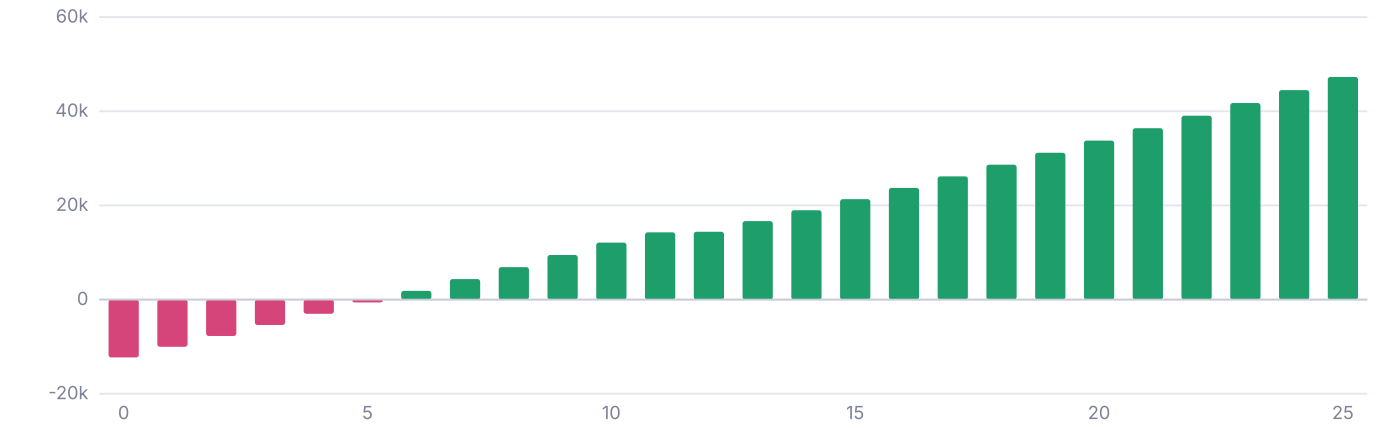
## 2. Energy, month by month




## 3. Your bills before and after

| MONTH | USE KWH | SOLAR KWH | USED ON SITE | EXPORTED | FROM GRID | BILL BEFORE | BILL AFTER | SAVING |
|---|---|---|---|---|---|---|---|---|
| Jan | 800 | 420 | 268 | 152 | 532 | RM 345 | RM 142 | RM 203 |
| Feb | 800 | 414 | 268 | 146 | 532 | RM 345 | RM 144 | RM 202 |
| Mar | 800 | 454 | 273 | 181 | 527 | RM 345 | RM 133 | RM 213 |
| Apr | 800 | 417 | 268 | 149 | 532 | RM 345 | RM 143 | RM 202 |
| May | 800 | 407 | 266 | 140 | 534 | RM 345 | RM 146 | RM 199 |
| Jun | 800 | 381 | 260 | 121 | 540 | RM 345 | RM 153 | RM 192 |
| Jul | 800 | 391 | 263 | 128 | 537 | RM 345 | RM 151 | RM 195 |
| Aug | 800 | 403 | 266 | 137 | 534 | RM 345 | RM 147 | RM 198 |
| Sep | 800 | 390 | 262 | 128 | 538 | RM 345 | RM 151 | RM 195 |
| Oct | 800 | 403 | 265 | 138 | 535 | RM 345 | RM 147 | RM 198 |
| Nov | 800 | 382 | 259 | 123 | 541 | RM 345 | RM 153 | RM 192 |
| Dec | 800 | 383 | 259 | 124 | 541 | RM 345 | RM 153 | RM 192 |
| **Year 1** | **9,600** | **4,845** | **3,178** | **1,667** | **6,422** | **RM 4,145** | **RM 1,763** | **RM 2,382** |

## 4. Return on investment




Still paying back  In profit  Year 0 = installation

| | |
|---|---|
| Investment | RM 12,285 |
| Savings over 25 years | RM 65,612 |
| Net present value | RM 17,704 at 6 % discount rate |
| Cost of solar power | RM 0.257 per kWh over the system life |

| YEAR | SOLAR KWH | BILL BEFORE | BILL AFTER | SAVINGS | O&M + PARTS | NET CASH | CUMULATIVE |
|---|---|---|---|---|---|---|---|
| 0 | | | | | | −RM 12,285 | −RM 12,285 |
| 1 | 4,845 | RM 4,145 | RM 1,763 | RM 2,382 | RM 123 | RM 2,259 | −RM 10,026 |
| 5 | 4,749 | RM 4,487 | RM 1,939 | RM 2,548 | RM 133 | RM 2,415 | −RM 602 |
| 10 | 4,631 | RM 4,954 | RM 2,181 | RM 2,773 | RM 147 | RM 2,626 | RM 12,102 |
| 15 | 4,516 | RM 5,470 | RM 2,956 | RM 2,514 | RM 162 | RM 2,352 | RM 21,331 |
| 20 | 4,405 | RM 6,039 | RM 3,278 | RM 2,761 | RM 179 | RM 2,582 | RM 33,773 |
| 25 | 4,296 | RM 6,667 | RM 3,674 | RM 2,993 | RM 198 | RM 2,796 | RM 47,292 |

## 5. Assumptions

Solar output is modelled from NASA POWER long-term monthly climate data using an hour-by-hour average-day simulation (HDKR sky model, NOCT cell temperature, 9 % DC and 1.5 % AC losses, inverter efficiency 97 %). Weather varies from year to year, so a single year can be several per cent above or below the estimate. Modules lose 0.5 % of output a year; the tariff rises 2 % a year; O&M costs 1 % of the system price a year. The inverter is replaced in year 12. Bills follow the TNB domestic tariff from 1 July 2025 (energy, capacity and network charges, retail charge, EEI rebate, KWTBB and SST). Under Solar ATAP, exported energy is credited at the energy charge rate up to the energy imported in the same month; unused credit lapses at month end. Credits are assumed for 10 years, after which the system is for self-consumption only. Tariff rules and rebates change; confirm current rates with TNB, SEDA and the Energy Commission. The design must be confirmed by a site survey, structural check and the final equipment datasheets, and installed by a registered contractor.

Prepared by Omer Yousif

Accepted by Sample client (landed home), date